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Enhanced Recovery of Local Anesthesia in Pediatric Patients: The Impact of Photobiomodulation on Reversing Anesthesia Effects

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Background: The split-mouth study design is used in oral health research and usually consists of 2 treatments randomly assigned to either the right or left side. This split-mouth study aimed to evaluate the efficacy of photobiomodulation on reversal of local anesthesia in 50 children aged 8-10 years.

Material/Methods: The study was conducted among 50 children: 27 girls and 23 boys, aged 8-10 years (mean age 9.38±1.15 years), who presented 2 carious maxillary permanent molars. One side was randomly assigned to the laser group (50 teeth), and the contralateral side to the control group (50 teeth). At the end of the treatment, photobiomodulation (PBM) was performed in the area of infiltration at 6 points, with 635 nm (25 children) (250 mW, 500 mW/cm^2, 15J) and 808 nm (25 children) (200 mW, 400 mW/cm^2, 12J) (SmartM PRO, Lasotronix, Poland). On the contralateral side, the laser's off-mode applicator was used. Anesthetic effect was evaluated by palpation test (soft tissues) and electrical test (dental pulp).

Results: After 15 minutes, in the laser group the return to normal sensations in the palpation test showed 88% (808 nm) and 68% (635 nm), and only 20% in the control group (P=0.04123). After 45 minutes, all the participants from the PBM group returned to normal sensations (P=0.21458). Dental pulp's excitability threshold was lower for both wavelengths compared to the control group (P=0.000001).

Conclusions: The identification of factors accelerating the recovery time to normal function, such as PBM, can be used as important data to eliminate self-injury secondary to local anesthesia (LA) in children.

Keywords: Anesthetics • Child • Low-Level Light Therapy

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Introduction

Pain and Local Anesthesia in Pediatric Dentistry

Pain and dental anxiety are a common reason for pediatric patients to present at the dental office or to avoid a visit [1]. Although local anesthesia (LA) is used as an important tool for reducing pain and anxiety, paradoxically, pain during needle insertion and loss of normal sensations secondary to anesthesia frequently causes non-compliance in children. It has been demonstrated that its effect lasts longer than the maximum time required for the most common dental procedures to be completed [2]. However, this prolonged soft-tissue anesthesia is inconvenient, unnecessary, or detrimental, especially in children, and a higher risk of self-injury has been reported, but there have been few studies on acceleration of recovery to normal sensation. The literature mostly recommends a pharmacologic method with phentolamine mesylate, but there are also promising new studies showing a positive effect of PBM [3,4]. The proposed mechanism of action of PBM, as an antagonist to vasoconstrictor, with regard to local anesthesia reversal, is that it increases local blood flow, resulting in the clearance of the anesthetic from the sub-mucosal tissues by the circulation [3]. The goal of this study was to evaluate the efficacy of PBM with 2 different wavelengths on the acceleration of reversal of the anesthetic effect of infiltration anesthesia in children as a self-injury management method.

The International Association for the Study of Pain defines pain as "an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage" [1]. Odontogenic pain related to early childhood caries with progressive pulpal involvement and dental trauma was reported at early stages of development among young children [1]. It is one of the prime motives for pediatric patients to attend dental care, especially emergency services [4]. Management of pain in pediatric dental care is a critical aspect of anxiety, often related to the induction of pain and exacerbation of pain perception in the future. Consequently, these young patients experience lower pain thresholds, which persist for longer and with exaggerated pain memory [5]. Thus, inadequate pain management stimulates negative responses and fear in children, which constitutes an obstacle for dentists to instill a positive attitude in pediatric patients. Hence, treating pediatric patients with minimal distress and pain has become a predominant objective for every pediatric dentist [6,7].

Self-Injurious Behavior Secondary to Local Anesthesia in Children

Adequate profound anesthesia is the foremost requirement for starting any dental procedure and relieves pain, especially when dealing with children [8]. The only drawback associated with local anesthesia (LA) is that occurrence of soft-tissue anesthesia can persist for 3 to 5 hours, while the procedure itself usually lasts for less than an hour [2]. Patients experience limitation of function in terms of speech difficulty, drinking, and eating. However, children can misinterpret altered sensation due to lack of normal proprioceptive feedback [9]. When long-acting local anesthetics are used, this uncomfortable altered sensation can lead to painful self-harm [6,10], such as biting the lips associated with the strange, new, and rare sensation of numbness or in the absence of pain sensation during repeated unintentional biting. When the anesthetic sensation is reverted to normal, painful wounds can appear, often linked with the adverse effect of the injection itself, not with a child's self-harm activity [8,9]. Thus, children who have challenging behavior and those with attention deficit hyperactivity disorder (ADHD) are at higher risk of self-injury, especially when undergoing the dental procedure under sedation [11].

It has been reported that a 13% of children receiving administration of LA in mandibles of experience injuries to the soft tissues of tongue, lips, or cheeks [12,13]. According to Rafique et al, 86% of patients receiving local anesthesia reported moderate dislike of postoperative numbness, and 14% had a strong dislike. Studies reported that this type of traumatic injury to the lips, cheeks, and tongue occur after inducing local anesthesia in 4-17.8% of patients, commonly between 0 and 12 years of age, with a higher prevalence in pre-school children [15,16]. In addition to the physical discomfort, some patients withdraw from public life while affected, refrain from eating and drinking, or accidentally injure themselves by biting their lips or tongue [17]. Early return of normal sensory feedback after using short-acting anesthetics and photobiomodulation can be helpful in preventing self-harm, especially among young children [18].

Photobiomodulation

Laser therapy, formerly known as low-level laser therapy, has many biological effects in different tissues. The use not only of coherent monochrome light sources (lasers) but also of non-coherent light sources such as LED is now referred to as photobiomodulation (PBM) [18].

PBM has been evaluated in many studies and gained increasing interest because of its effectiveness in pain reduction by inhibition of neural function and for pain prevention in various fields of dentistry [3,18,19].

It is reported that the biological effect of PBM occurs by the absorption of light through receptors (chromophores) present in tissues. The most important chromophore in mammalian tissues is cytochrome C oxidase (CCO) [3,18,20]. The hypothesized mechanism of action is that inhibitory nitric oxide (NO) can be dissociated from CCO, thus restoring electron

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transport, and increasing mitochondrial membrane potential. It has been demonstrated PBM produces free NO in the irradiated tissues. Free NO functions as a powerful vasodilator and an important cellular signaling molecule that is crucial in various physiological processes [21-23].Vasodilation is a mechanism of action of phentolamine mesylate, a medication used in reversal of soft-tissue anesthesia as an injection using the same method (infiltration, block) as a previous anesthetic injection [13,14]. PBM is a non-invasive therapeutic approach widely applied for medical purposes, including pain management in pediatric dentistry. Some patients receive it as a separate treatment, while others receive it combined with a conventional pharmacologic approach [18,19].

Studies conducted to reverse the effect of local anesthesia using PBM therapy are encouraging, but still very limited. Maegawa et al and Kubota showed the potential of 830-nm PBM to improve the blood flow and increase microcirculation in an animal model [20,21]. A review of the literature showed that, in an in vivo model, this therapy causes an increase in blood microcirculation by a reduction in isometric tension of the vascular smooth muscle. The administration of percutaneous PBM enabled reversal of histamine-induced spasm in atherosclerosis by relaxing the vascular smooth muscle [22]. On the basis of this mechanism of action from an animal model Seraj et al conducted a study including 34 children aged 4-8 years to reverse soft-tissue local anesthesia using 810-nm PBM. They showed a reduction of 43 min in the reversal duration of soft-tissue local anesthesia in comparison to those without PBM [24].

There has been only 1 preliminary study on the effect of PBM on soft-tissue local anesthesia reversal, so our study aimed to compare soft-tissue local anesthesia reversal at 2 different wavelengths, thus helping to reduce the consequences of prolonged numbness, as PBM might be helpful in increasing the depth of anesthesia and decreasing the prolonged unnecessary anesthetic effect [25]. On the basis of these findings, implementation of photobiomodulation as an adjunct to local anesthesia might increase the microcirculation and accelerate elimination of anesthetic drug, reduce the time of unnecessary anesthesia or prolonged numbness associated with high risk of self-injury in young children undergoing local anesthesia for dental treatment.

Thus, the main aim of this study was to evaluate the efficiency and acceptance of anesthesia reversal induced by PBM using a 635-nm and 808-nm wavelength laser in comparison to standard time of local anesthesia used for restorative dentistry procedures in children. We hypothesized that PBM leads to changes in microcirculation, as was showed in previous studies, and reduction in the reversal time duration of soft tissue and pulp local anesthesia, resulting in less risk of self-injury in children.

The study involved children aged 8-10 years who qualified for caries treatment in molars with the use of infiltration anesthesia on both the right and left sides of the maxilla.

The study population was chosen according to the following factors: self-injury incidence, caries prevalence in first permanent molars, age at which root development enabled electrical pulp testing, and cooperation level of subjects in helping them to understand the procedure and evaluating the results of palpation and pulp testing.

The split-mouth design, which divides the child's dentition into right and left halves, is commonly used in clinical research on pediatric dentistry. Two different treatment modalities are randomly assigned to a side for further outcome assessment, and each participant acts as their own control. This study design uses within-patient rather than between-patient comparisons for results analysis. Thus, advantages of split-mouth study are decrease in inter-subject variability and increased study accuracy and power. However, the main disadvantage with this design is the potential contamination of the obtained treatment effect from one side to the other, known as the "carry-across effect." To diminish this problem, in our study we scheduled a second appointment after 10 days. Also, it is easier for a child to cooperate with a 45-min appointment, which corresponds to the length of school lessons. The split-mouth study design is not recommended when the evaluated oral disease is not symmetrically distributed in the mouth of children. In our study, an inclusion criterion was having the same number of decayed tissues (4-5 according to ICDAS) in both molars. To increase validity and reliability of obtained data from split-mouth design studies, we followed recommended evaluation and adequate selection of the inclusion and exclusion criteria.

The protocol of our study was designed to compare the duration of soft-tissue anesthesia as one session of laser therapy at one side and the other side without laser as a control-sham laser.

The predictor variable of our research was the photobiomodulation skills in the aspect of discomfort secondary to local anesthesia. The primary endpoint was the time needed for the reversal of soft tissues and pulpal anesthesia after irradiation with 635 nm and 808 nm. The secondary endpoint was the occurrence of possible adverse effects due to photobiomodulation used after infiltration anesthesia.

We explored the increased discomfort secondary to local anesthesia in young patients when the risk of self-injury is relatively high and the only existing protocol is injection with phentolamine mesylate, which with anxious children is often difficult to perform. Non-invasive PBM increases local blood flow, resulting in clearance of local anesthetic from the sub-mucosal

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Table 1. The Frankl behavior rating scale.

Categories of behavior
Rating 1: Definitely negative: Refusal of treatment, crying forcefully, fearfulness, or any other overt evidence of extreme negativism
Rating 2: Negative: Reluctance to accept treatment, uncooperative behavior, some evidence of a negative attitude but not pronounced (i.e., sullen, withdrawn)
Rating 3: Positive: Acceptance of treatment, at times cautious, willingness to comply with the dentist, at times with reservation but follows the dentist's directions cooperatively
Rating 4: Definitely positive: Good rapport with the dentist, interested in the dental procedures, laughing and enjoying the situation

A four-point scale with 2 degrees of positive behavior and 2 degrees of negative behavior. Inclusion criteria for the study was cooperation rate 3-4.

tissues to the circulation, accelerating the return of normal sensations in the anesthetized area.

Material and Methods

Ethics Approval

This research was carried out in accordance with the Helsinki Declaration of the World Medical Association.

The trial was designed as a randomized and controlled test. The approval of the Local Ethics Committee of Wrocław Medical University, Faculty of Dentistry was obtained before the study begun (permission number: KB235/2021) and informed consent was obtained from all participating subjects' parents. In research involving children under the age of 18, who cannot give consent for medical procedures and treatments, parents are typically the primary decision-makers for their children. The parent giving consent must be deemed competent and must be able to understand the information being presented.

However, in pediatrics, children who are old enough to understand medical discussions may be asked to give assent for care. Consent means that the child is agreeing to the treatment or procedure. Children can also refuse consent, which means they do not agree to participate. The age at which consent is requested varies and can be as young as 7 years old [2]. Consent is not required by law, but in our study we required children who were developmentally and cognitively able to participate in decision making.

The study protocol was explained to all eligible children and their parents, and informed consent forms were obtained before beginning the study. All patients' parents/caregivers were informed in detail about the procedure and expected benefits from the implementation of PBM secondary to local anesthesia. It was also explained to them that the only failure of PBM is the lack of effect, meaning that the anesthesia effect would not be shortened. We only included children whose parents understood the aim of the study and signed informed consent. Typically, both parents need to provide consent for a child to participate in a clinical trial.

Subjects

The analysis concerned the results of photobiomodulation following infiltration anesthesia of children aged 8-10 years who had this treatment performed at the Department of Pediatric Dentistry at Poznań University. The mean age of the participants was 9.38±1.15 years (range from 8 to 10 years). The distribution of genders was 54% female (27 individuals) and 46% male (23 individuals). Age and gender were not significantly different between the 2 study groups.

A total of 55 patients were selected for this study from the pool of children referred for routine treatment. Then, we identified a group of 50 children who fulfilled all the inclusion criteria. We enrolled 50 children who presented 2 maxillary permanent molars in right and left quadrants (n=100 teeth). We used SPSS software version 25 with paired means analysis and running tests (α=0.05 and power=95% and effect size=0.5). The minimum sample size required for this study was determined on 25 subjects in each laser and sham laser group.

At the first appointment ("1") patients were divided into 2 groups: "1A" 25 children (n=25 molars) received irradiation 635 nm, and "1B" 25 children (n=25 molars) received irradiation with 808 nm.

At the second appointment ("2"), after 10 days, the same children (25 molars in each group) were treated without irradiation as a control group (2A and 2B).

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The research involved children with a 3-4 degree of cooperation according to the Frankl scale. The Frankl classification method, as seen in **Table 1**, is often considered the criterion standard in clinical rating scales, mainly as a result of its wide usage and acceptance in pediatric dentistry research [23]. All the subjects of this study were treated by the same pediatric dentistry specialist.

Inclusion Criteria

The inclusion criteria for the study were as follows: children aged 8 to 10 years old, who are able to understand and respond to the test of sensation from soft tissue/dental pulp, with a Frankl cooperation scale level of 3-4. For each child, a written informed consent form was signed by the parents/caregivers. Patients had undergone a dental check-up before the clinical trial. Each patient had at least 2 similar contralateral dental caries lesions (4-5 according to ICDAS) in the same jaw/maxilla (for split-mouth design analysis), assessed in radiographic evaluations and clinical examinations. All the teeth were vital and showed a normal response to the thermic vitality test, and had complete root formation and no periapical lesions detected in radiographs. The treatment was followed by infiltration anesthesia with articaine (Citocartin 200 with 1: 200 000 epinephrine, Molteni Dental, Italy) using a computer-controlled delivery system (Calaject, Ronvig, Denmark) program 1 infiltration. We enrolled patients who met the following inclusion criteria: no systemic diseases, no allergy to the anesthetics and sulfite, no use of analgesic drugs before treatment, no contraindication for local anesthesia/no systemic diseases, no use of anti-inflammatory drugs 24 h before the examination/treatment, no use of antibiotics in the previous 6 months, no uncompensated diabetes or uncontrolled periodontal disease, and no history of radiotherapy or chemotherapy.

Exclusion Criteria

We excluded patients with a history of any medication that might affect anesthetic assessment and those who had pathological lesions in the injection area. We excluded children who required surgical procedures where postoperative soft-tissue anesthesia may be preferred to avoid post-procedural pain. We excluded children who would be unable to score correctly on the palpation and electrical test, or if cooperation level was 1-2 on the Frankl scale. We excluded children if the treatments on both sides had unequal amounts of decay (or more than 5 ICDAS). We excluded children who had not achieved profound cheek numbness, requiring additional anesthesia, or those who could not learn to distinguish the numb anesthetized side from the non-anesthetized side, but they were still offered the intended dental procedure in the same visit. We excluded children with previous negative experiences with anesthesia/dentistry or subjects, or who did not return/cooperate for the second treatment.

Randomization

Patients who met the inclusion criteria and their parents/caregivers were further asked to fill in the questionnaire (eg, demographic data, age, sex). The study used a split-mouth design, and each patient was assigned by simple randomization (opening a sealed envelope) to one side of the maxilla to the laser group and the contralateral side was assigned to the sham laser group.

The randomization was based in 4 opaque sealed envelopes to determine which group (PBM or sham) would be the first treatment and at which side of the mouth (left or right) the treatment should be started. The participants selected an envelope and the treatments were applied. The operator, who was also the examiner, was not blinded to the treatment as the test group received a PBM (635 nm or 808 nm), while the control group did not receive any irradiation.

Assessment Methods

Evaluation of anesthetic effect (primary endpoints) included: assessment of the pulpal anesthesia by electrical pulp test (dental pulp excitability threshold [VT]) using a Vitality Scanner (Sybron Endo, Kerr, USA). The examination was performed 3 times: before anesthesia, before laser irradiation, and after PBM (45 min after administration of anesthesia). The evaluation of anesthetized soft tissues was done by palpation testing at 15, 30, and 45 min after anesthesia (results: 0 – no reaction, 1 – normal sense of touch).

Study Protocol

After the dental clinical and radiological examination, all the children were informed about tests and methods of evaluation of recovery of sensation of touch and pain. One pediatric dentist explained the scales used in this study to the pediatric patients in age-appropriate terms and also performed all clinical procedures.

The study protocol included a pilot study what included group of 12 patients (7 girls and 5 boys) who were divided into 2 laser groups (635 nm and 808 nm). Patients were informed about tests (cold test, electrical test, and palpation) and the possible sensations, as well as how to report them.

Prior to initiation of the study, the operator/investigator and a patient were trained to assess cheek numbness using index finger palpation and tapping with comparison to the non-anesthetized side, and to evaluate sensations from electrical pulp test and thermal test in 12 patients not involved in the study.

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Reliability of participants was tested with a sham test, by using a cotton pellet sprayed with an empty bottle of Green Endo-Ice® (Coltene Whaledent, Cuyahoga Falls, OH, USA) placed on a caries-free control tooth. Participants who responded positively to the sham test were excluded from the study. A pulp electrical test was performed on the control mandibular molars. Patients were educated about the expected sensations and the method of recording by raising a hand. In case of lack of sensation at the value 80 or hypersensitivity (lower pain threshold), subjects were excluded from the study. A control tooth (mandibular molar) was tested also by an inactivated electric pulp tester to test the reliability of the subject. If the subject responded positively to an inactivated pulp tester, then they were considered unreliable and were excluded from the study.

As a calibration procedure, children were taught about the experiment, about sensations accompanying infiltration anesthesia in maxillary molars area, and how to evaluate and record them. After injection, the palpation technique was been used to assess soft-tissue anesthesia. We also excluded children who did not understand the procedure or had lack of effect or needed an additional injection.

Preoperative Phase

The teeth chosen for the experiment were the first maxillary molars. According to the literature, those are the most common teeth affected by caries, and at the age of 8-10 years, infiltration is a recommended technique of LA [19]. A visual, clinical, and OPG (oral pantomogram) examination was conducted to ensure that all teeth were free of large restorations or periodontal disease and that none had a history of trauma or sensitivity, and had only medium caries without pulp involvement. Before the experimental study, at both appointments, the chosen maxillary molars and the contralateral mandibular molar (control) were tested 2 times with an electric pulp tester (Vitality Scanner, Sybron Endo, Kerr, USA) to ensure tooth vitality and obtain baseline information. Electric pulp testing (EPT) is a commonly used method to test pulp sensibility. The sensation the patient feels when an electric current is passed through the tooth structure is the result of direct stimulation of the pulp nerve fibers. The principle of the EPT, whether it be a type that measures voltage or current, is to raise the electrical potential through the enamel and dentine into the pulp to provoke a measurable response from the pulp. Electrical stimulation of nerves within the pulp depends on the rate of current increase, and its strength (voltage and current), duration, and frequency. Each stimulation should have a certain minimum intensity/strength to excite, and this is called the threshold stimulus (VT). The vitality scanner has a rheostat that shows the relative amount of current applied on various scales, ranging from 1 to 80. The current flow should be increased slowly to allow the patient enough time to respond before warmth or a tingling sensation becomes painful. The sensation may be noted by the patient as a tingling sensation, warm, stinging, fullness, or heat. A response to an electrical test does not provide any information about the health status of the pulp, its circulation, or its integrity. It only indicates that some sensory fibers are present within the pulp tissue and are capable of responding to the stimulus.

The teeth included into the study were isolated with cotton rolls and dried with an air syringe. Toothpaste was applied to the probe tip, which was placed in the middle third of the buccal or facial surface of the tooth being tested. Patients were instructed about the analytical procedure, expected sensations, and methods of report (raising a hand). The current rate was set at 25 seconds to increase from no output (0) to the maximum output (80). No response from the subject at the maximum output (80 reading) of the pulp tester was used as the criterion for exclusion from the study. The value at the initial sensation was recorded (VT1). The current rate was set at 25 seconds to increase from no output (0) to the maximum output (80).

Operative Phase – Local Anesthesia

According to the randomly chosen side, in the laser procedure for the first session, the maxillary infiltration was administered using 1 ml of 4% articaine (Citocartin 200 with 1: 200,000 epinephrine, Molteni Dental, Italy) with a standard a 27-gauge 1½-inch needle (Becton Dickinson and Co, Franklin Lakes, NJ, USA) and a computer-controlled delivery system Calaject (Ronvig, Denmark). Dental injections were performed by the same pediatric dentist using the Calaject system program 1 dedicated to infiltration anesthesia. The system controls the flow rate of anesthetic, which ensures a smooth and gentle flow of anesthesia. Due to the physiological slow speed of liquid deposition, injection is almost pain-free and is well accepted by patients.

Before administering the injection, the mucosa was dried, and a 27-gauge needle with a first drop of anesthetic was gently placed onto the alveolar mucosa (topical anesthesia) for 20 s. Then, after pressing the pedal, the needle insertion phase started with very slow deposition and advanced within 2-3 seconds until the needle was estimated to be at the target site (needle placement phase). The anesthetic solution was deposited with slow speed for 20s and continued during a period of 1 min (solution deposition phase). After deposition of the anesthetic solution at the target site, each subject was asked if they were experiencing cheek numbness. After a 2-min waiting period, cheek numbness was confirmed, and the tooth was tested for preoperative pulpal anesthesia using the Green Endo-Ice® cold test.

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Before the experiment, each patient received detailed trained about testing methods, possible sensations, and how to express them. Two minutes after injection, a microbrush soaked with cold spray ("freezing" effect) was placed on the healthy enamel on the buccal surface of the tooth. If there was no reaction, the tooth was included into the study and qualify for further examination. In a case of sensitivity to cold test in 2 subsequent tests (every 30 s), the patient was excluded from the study because of the need for additional anesthesia.

The success of preoperative pulpal anesthesia was defined as 2 consecutive negative responses to cold testing. Then, the cavity preparation was done and the filling was placed. We excluded patients who had pain or sensitivity during cavity preparation and needed a second injection.

Postoperative Phase

The electrical pulp test was performed when the cavity preparation was finished and the filling was placed. The value at the initial sensation was recorded (VT2). Then, irradiation with laser at 635 and 808 nm was performed according to the randomized group. In the sham laser group, the laser tip in off-mode was placed at the same points. In the second session, after 10 days, the opposite-side molar was treated and results were recorded in the patient's chart. The third electrical pulp measurements were done 45 min later in the 635-nm and 808-nm groups. The same measurements were done in the control group (VT3). Subjects in both groups (irradiated and sham laser) also palpated their cheeks every 15 min after irradiation (at 15, 30, and 45 min) after the procedure was finished and recorded when a return to normal sensation (absence of numbness and pins-and-needles sensations) occurred. Children were instructed how to accurately palpate the cheek. These recordings were used to calculate the efficacy of reversal of soft-tissue anesthesia by using photobiomodulation. The time to recovery of normal soft tissue was the number of minutes elapsed from irradiation to the first of 2 consecutive times at which the patient reported a normal sensation of the soft tissue.

Laser Application

Before the experiment and patient participation, the laser was conveniently situated, and warning notices for the laser use were clearly and plainly displayed. During the PBM and placebo PBM applications, wavelength-specific protective eyeglasses (Doris CTL 2109S, Poland) were used by both the operator and the patient. To ensure the efficiency of the device, the output power was measured with a power meter 3 times throughout the research (Vega Power Meter; Ophir Photonics, 3050 North, 300 West North Logan, USA).

On the first session, 45 min after the injection and when treatment was finished, patients assigned to the laser group ("1") received diode laser irradiation (SmartM, Lasotronix, Poland): group 1A with a wavelength 635 nm and group 1B with 808 nm.

The laser applicator (8-mm diameter) was placed directly on the mucosa and skin (contact technique). The points of irradiation were located at the injection site: 2 points on the buccal (mesial) side and 2 points on the palate (distal) side, 1 point in the oral vestibule on the cheek, and 1 point on the cheek skin.

According to the safety measures during the irradiation procedure, both patient and operator wore appropriate safety goggles. Laser parameters and irradiation protocol are shown in the **Tables 2 and 3**, respectively.

Assessment of Soft Tissues Sensations After Anesthesia

Palpation technique was used for evaluation of anesthesia effect on soft tissues at 15, 30, and 45 min after irradiation until the first time of reversal of anesthesia and return of normal sensation during palpation was achieved.

According to the literature and median time of local anesthesia sensations experienced by the patients, the time intervals for evaluation of soft tissue and pulpal anesthesia were chosen as 15 min and repeated 3 times; 45 min after anesthesia corresponds to a median time of pediatric dentistry procedure, and after 45 min of observation, most patients who received PBM had normal sensations from the soft tissues and were at less risk of self-injury [24].

The patient's response of "yes" or "no" was recorded as indicative of feeling or not feeling the stimulus. They were instructed that this procedure would be repeated and that every time they should report either feeling normal or numb on the anesthetized side in comparison to the non-anesthetized side. The time of returning of normal sensations or persisted anesthesia were recorded in the patient's record.

The same operator examined all patients during the observational period for a self-inflicted injury. It was recorded 2 times for each patient – at the time of dismissal and after 24 h, by telephone call, to reveal the time at which the injury, if any, had occurred. Self-inflicted injury was documented in the patient's chart in the form of redness, hematoma, swelling, or ulcer, each recorded separately.

Statistical Analysis

We conducted statistical analyses using Statistica® Version 13.5.0 software for Windows (TIBCO Software, Inc., Palo Alto, CA, USA) and PQStat 1.8.0.414 software (PQStat software;

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Table 2. Laser parameters in photobiomodulation group (G1A – 635 nm and G1B – 808 nm).

Parameters	635 nm laser Group G1A	808 nm laser Group G1B
Power (mW)	250	200
Applicator diameter	8mm	8mm
Beam spot size at target (cm^2)	0.5	0.5
Irradiance at target (mW/cm^2)	500	400
Exposure duration (s)	10 per point	10 per point
Radiant exposure (J/cm^2)	5	4
Radiant energy (J)	2.5	2
Number of points irradiated	6	6
Area irradiated (cm^2)	3	3
Application technique	Contact	Contact
Number and frequency of treatment sessions	1	1
Total radiant energy (J)	15	12
Operating mode	Continuous mode	Continuous mode

Table 3. Photobiomodulation irradiation protocol.

Laser procedures	Photobiomodulation group G1A (635 nm)and G1B (808 nm)	Control group G2A and G2B (sham)
After treatment	First session: One side Group 1A 635 nm irradiation Group 1B 808 nm irradiation /according to laser parameters Table 2/	Second session: /After 10 days/ Opposite side molars Group 2A and 2B Laser OFF-mode The same points

Poznań, Poland). Our approach involved comparing results both before and after laser administration, taking into account different laser wavelengths. Prior to analysis, the normality of the data was assessed using the Shapiro-Wilks test. Given the non-parametric distribution observed in the data, we used the 2-sided Mann-Whitney U and Wilcoxon tests. For the pre-laser vs post-laser application case-control study, the Kruskal-Wallis test with Dunn's post hoc test were utilized. Descriptive statistics for experimental results were presented based on the distribution of variables, with the median [lower-upper quartiles] (Me [Q1-Q3]) reported. In addressing the issue of multiple comparisons, we applied the Bonferroni correction method. This approach effectively controlled the familywise error rate by adjusting the significance level for each individual comparison. Nominal data were analyzed using the Bowker-McNemar and Q-Cochrane tests, and the outcomes were expressed in terms of the number of observations. Statistical significance was defined at a corrected *P* value of <0.05, ensuring a robust control for type I errors in the context of multiple comparisons.

Results

The study was performed in the Department of Pediatric Dentistry of Poznań Medical University, Poland from March 2021 to September 2021 (recruitment and investigation), and the follow-up ended in December 2021. Fifty children (n=50) – 54% female (27 individuals) and 46% male (23 individuals) – participated in this split-mouth study. Patients were all of the same ethnicity, and parents/caregivers declared the children had dental checkups every 6 months. The mean age of the participants was 9.38±1.15 years (range from 8 to 10 years). Age and sex were not significantly different between the 2 study groups.

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In total, 100 maxillary permanent molars were included in the study. During the first session, 50 maxillary molars, 25 in each laser group, were evaluated according to the randomization procedure. After 10 days, the 25 opposite-side maxillary molars were analysed in each control-sham laser group. The palpation technique was used for evaluation of the sensations from soft tissues after anesthesia (at 15, 30, and 45 min), and pulpal anesthesia was measured by electrical test before and after irradiation (group 1) or placement of laser applicator in "off" mode (group 2).

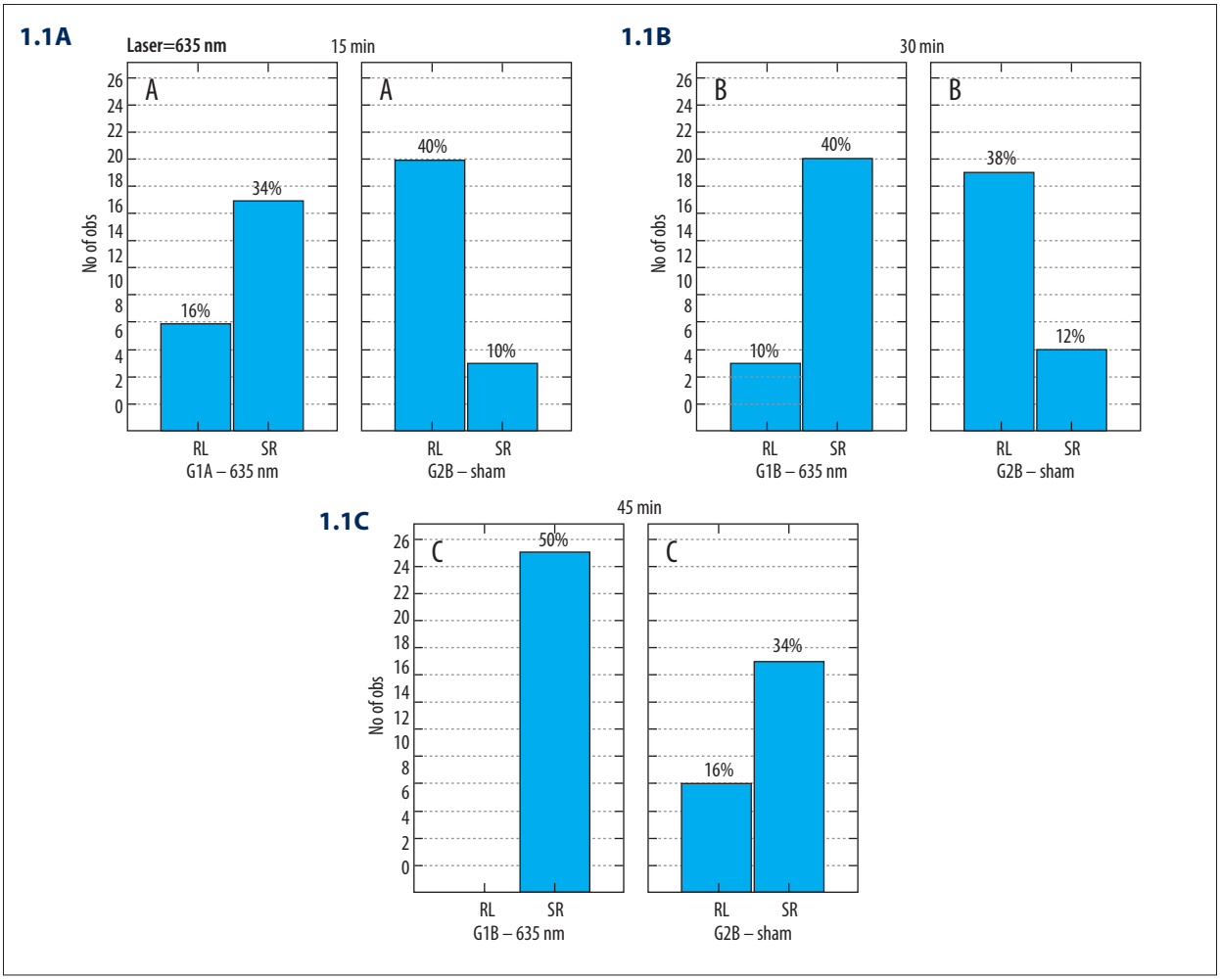
Duration of Soft-Tissue Anesthesia

The Bowker-McNemar and Q-Cochrane tests were used to study the degree and time of reversal of soft-tissue anesthesia in the laser group (G1A, G1B) and control group (G2A, G2B). **Figure 1** shows the time of anesthesia sensations and return to normal function expressed in minutes in both groups.

The intra-group comparison of the duration of anesthesia showed a significant difference between the laser group and the control group ($P<0.05$). The study showed reduced anesthesia time to about 45 min after laser irradiation in all subjects (G1A and G1B). Furthermore, for the 808-nm wavelength, the return to normal sensation was after 30 min, which was faster than with 635 nm, but the difference was not statistically significant ($P>0.05$) (**Figure 1**).

Pulpal Anesthesia Assessed by Using Electrical Test

Regardless of the wavelength of the laser applied, we observed differences in the dental pulp's excitability threshold between pre- and post-laser application. After laser application, the measured vitality test/dental pulp's excitability threshold was significantly lower, which corresponded to recovery of pulp anesthesia ($P<0.001$). We observed differences in the threshold of the dental pulp excitability between the control-sham laser group and post-laser application in the case of the 635-nm laser (1A group) ($P<0.001$) and 808 nm (1B group) ($P<0.000$). The median ranks for 635 nm and 808 nm were 62 [55-70] and 61 [50-70], respectively. In examined irradiated cases, the median level of the pulp vitality post-laser in analyzed wavelengths corresponded to return of normal pulp sensitivity and



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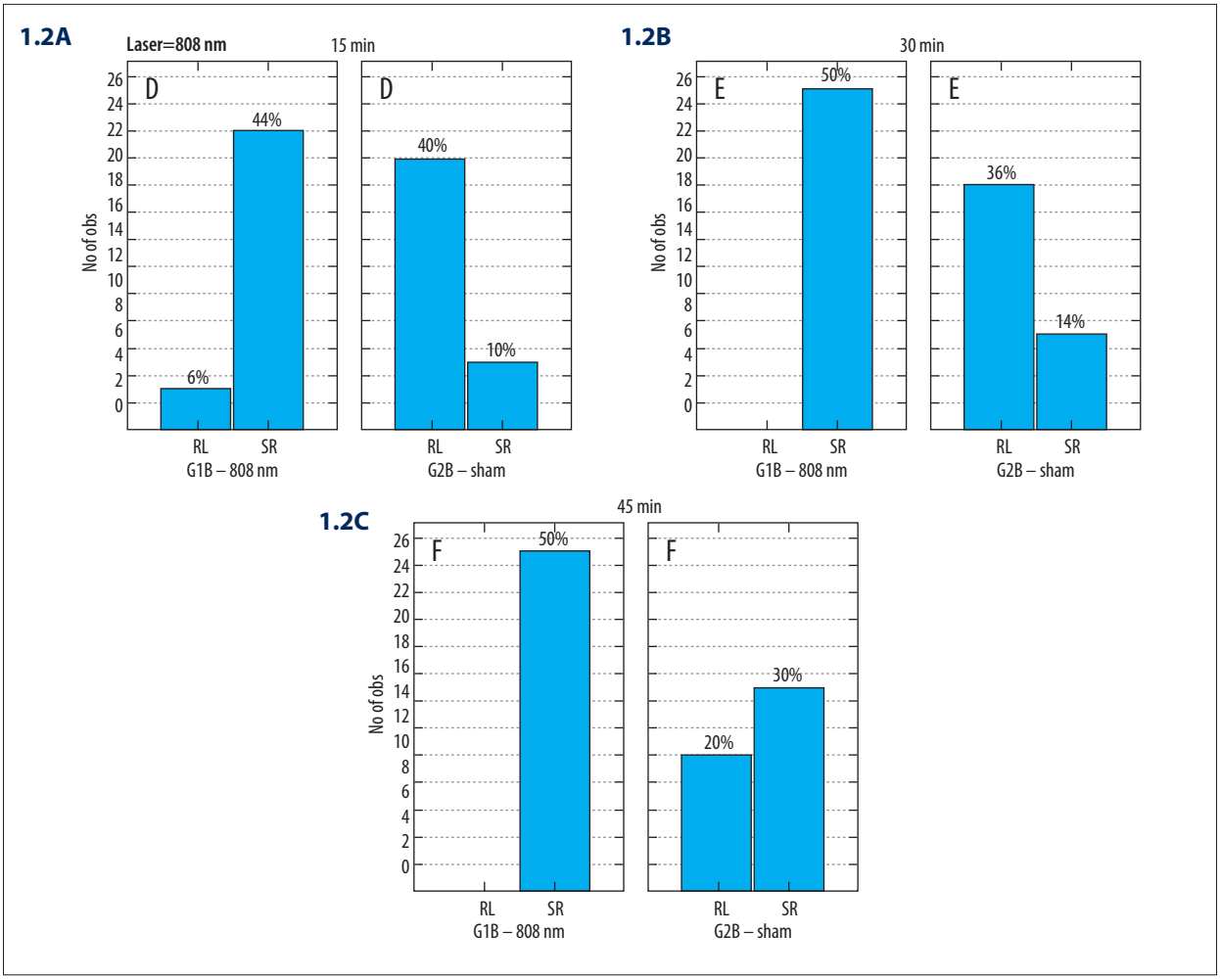


Figure 1. Changes in soft-tissue anesthesia sensations with time in laser group where photobiomodulation (PBM) was performed (G1A – 635 nm and G1B – 808 nm) in comparison to control group (G2 sham laser). The results were expressed in the number of cases (No of obs) and a percentage (%) of the total patients' number per group, NS – not significant ($P>0.05$); NA – P value not available. The same capital letters (A,B,C,D,E,F) indicate a significant difference between groups. *The statistical analysis and graphs in this figure were produced using Statistica® Version 13.5.0 software for Windows (TIBCO Software, Inc., Palo Alto, CA) and PQStat 1.8.0.414 software (PQStat software; Poznań, Poland).* No of obs – number of observations (number of patients);% of all cases; RL – lack of reaction = anesthesia; SR – return of normal sensation; NS – not significant (P>0.05); NA – P value not available (due to lack of cases in RL group); min – minutes; nm – nanometers (**1.1A**) After 15 min PBM with 635-nm laser in comparison to sham laser. (**1.1B**) After 30 min PBM with 635-nm laser in comparison to sham laser. (**1.1C**) After 45 min PBM with 635-nm laser in comparison to sham laser. (**1.2A**) After 15 min PBM with 808-nm laser in comparison to sham laser. (**1.2B**) After 30 min PBM with 808-nm laser in comparison to sham laser. (**1.2C**) After 45 minutes PBM with 808-nm laser in comparison to sham laser.

68 [65-70] means still persistent pulpal anesthesia in the sham laser group (**Figure 2**).

After irradiation (at 635 nm and 808 nm), faster recovery of pulp vitality function was observed as measured by electrical test. The pulp excitability threshold in the laser groups had lower scores than the sham groups was reaching, which means the patients experienced sensation from the dental pulp with electrical stimulation, while the anesthetic effect persisted in sham laser group (**Figure 2**).

It has been reported in the literature that after local anesthesia the pulp tissue reverts to normal function first, in comparison to the supporting tissues (lips, cheeks, tongue), which remain anesthetized for several hours [26,27]. It has several clinical implications. Although after biological vital pulp therapy patients could benefit from prolonged pulpal anesthesia, only after surgical interventions prolonged soft tissues anesthesia might be favorable. In other cases,especially in children and disabled patients it might result self-injury to the numb lips, cheeks and tongue.

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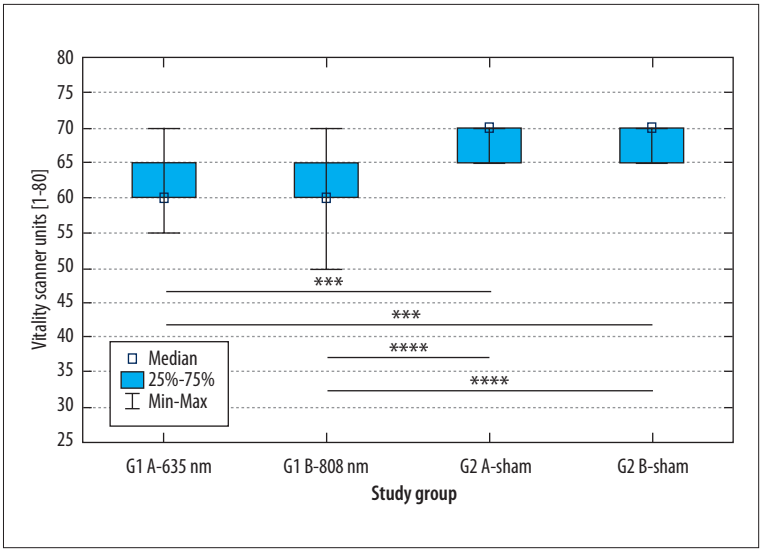


Figure 2. Changes in dental pulp sensations secondary to anesthesia with time in photobiomodulation group (G1A 635-nm, G1B 808-nm) in comparison to control (sham laser G2 – A and B). Data are presented as box-whiskers plot (median, lower-upper quartiles [Q1-Q]) of the dental pulp excitability level before and secondary to anesthesia and photobiomodulation (PBM). Horizontal lines indicate the differences between the groups. Statistical significance was defined at a *P* value <0.05 and presented as *. *The statistical analysis and graphs in this figure were produced using Statistica® Version 13.5.0 software for Windows (TIBCO Software, Inc., Palo Alto, CA) and PQStat 1.8.0.414 software (PQStat software; Poznań, Poland).*

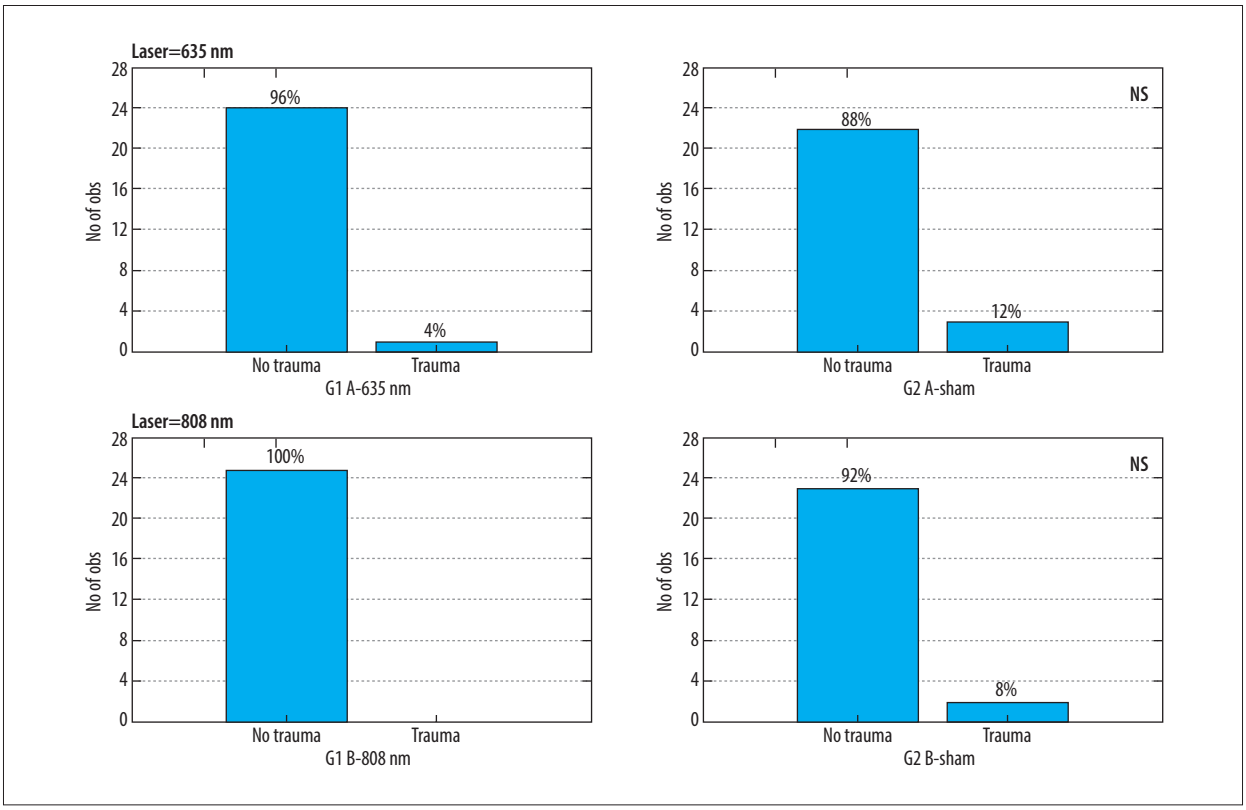


Figure 3. Distribution of the self-inflicted soft-tissue trauma 24 h after the appointment in laser group (635 nm and 808 nm) in comparison to control group (sham laser). The results were expressed in the number of cases (No of obs) and a percentage (%) of the total patients' number per group, NS – not significant (*P*>0.05); NA – *P* value not available. *The statistical analysis and graphs in this figure were produced using Statistica® Version 13.5.0 software for Windows (TIBCO Software, Inc., Palo Alto, CA) and PQStat 1.8.0.414 software (PQStat software; Poznań, Poland).* No of obs – number of observations (number of patients); NS – not significant (P>0.05); NA – p-value not available (due to insufficient number of cases);% – percentage of cases; G1A – first appointment with 635-nm photobiomodulation; G2A – second appointment, after 10 days, sham – control group (without laser irradiation), G1B – first appointment with 808-nm laser photobiomodulation; G2B – second appointment, after 10 days, sham – control group (without laser irradiation).

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Table 4. Duration of action for several local anesthetic agents used in dentistry.

Local anesthetic	Approximate duration of pulpal anesthesia [minutes]	Approximate duration of soft tissue anesthesia [minutes]
Bupivacaine hydrochloride 0.5% with epinephrine 1: 200,000	>90	240-720
Lidocaine hydrochloride 2% without vasoconstrictor	<10	30-45
2% with epinephrine 1: 50,000	60	180-300
2% with epinephrine 1: 100,000	60	180-300
Mepivacaine hydrochloride 3% without vasoconstrictor	5-10	90-120
2% with levonordefrin 1: 20,000	60	180-300
Prilocaine hydrochloride 4% without vasoconstrictor	40-60	120-240
Articaine 4% with epinephrine 1: 100,000	60-90	180-480
Articaine 4% with epinephrine 1: 200,000	45-60	180-240

The Soft-Tissue Self-Injury

No statistically significant differences were found between the laser and sham groups related to soft-tissue self-injury. In both control groups, there were 4 more cases of self-injuries due to cheek biting after the administration of infiltration anesthesia for treating caries, which is considered important from a clinical point of view. Although after 45 min in the palpation test, children reported normal sensations, on the phone call the next day, 1 case of self- injury was reported in the irradiated group and 5 cases in the sham laser group. These results suggest that the degree and duration of soft-tissue anesthesia is associated with increased risk of self-injury by unintentional biting of the lips, cheeks, or tongue, especially in young patients (**Figure 3**).

Discussion

The potential of PBM to return patients' soft tissues to normal sensation and to avoid self-inflicted injury are highly desirable in the pediatric dentistry and would be perceived as advantageous by many pediatric patients and their parents and/or guardians. The results revealed in our study establish a promising clinical value for PBM in means of reducing the severity of postoperative discomfort and in accelerating recovery to normal function from local anesthesia in pediatric patients after dental procedures.

There is a need for development of new technologies which can help to deal with dental pain [28,29]. Adult patients often feel uncomfortable and expect deep and long-acting anesthesia to protect them from pain during and after the dental procedure. In most cases, they expect the same for their children [26]. In the present study, we used 4% articaine with adrenaline 1: 200 000 because of its short time of action, but it is adequate for most procedures in pediatric dentistry [30] (**Table 4**). The superiority of articaine may be related to its intramolecular hydrogen bonding in thiophene rings, allowing for better bone penetration [11,25]. The volume of anesthetic solution may also affect anesthetic success; however, there is no consensus on the appropriate volume of articaine for effective pulpal anesthesia with LA [25]. Results from the present study show that using three-fourths of a cartridge of 4% articaine with 1: 200 000 adrenaline with slow deposition using computer system program 1 increased the preoperative pulpal anesthetic success in all subjects. Long-acting local anesthetics also have higher neuronal toxicity and are not recommended for children [24]. The duration of soft-tissue numbness is considerably longer than that of pulpal anesthesia and longer than the duration of the typical dental appointment [27]. In addition, longer periods of anesthesia not only affect eating, drinking, talking, and smiling but can result in greater risk of developing injury to the lips or cheek due to biting numb tissues, especially in young children and disabled patients [28].

After 24 h, 1 case of trauma was reported in the laser group and 5 cases were reported in the sham laser group. At the end of the appointment, in the irradiated group, all the children answered positively to the palpation test, but in the control group, only 64% answered positively, which means 36% still

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sensed soft-tissue anesthesia, which might explain the traumatic lesion. A prospective study conducted by College et al on 320 children showed that self-inflicted injury occurred according to the patient' age – the younger the child, the higher risk – showing that 18% of children younger than 4 years old, 16% of children aged 4-7, and 13% of children aged 8-11 treated by injection to the inferior alveolar nerve block had some degree of soft-tissue damage [28,29].

The literature contains few studies on the efficacy of different methods used to alleviate unpleasant secondary sensations to local anesthesia. Two different studies by Ram et al assessed the efficacy of using ice or popsicles in reducing discomfort or self-inflicted biting after receiving local anesthesia. They found that the children were less restless, with less self-inflicted biting [25].

PBM therapy has been widely applied in dentistry in the last 2 decades [30-38]. The present preliminary clinical trial study was designed to evaluate the efficacy of photobiomodulation in reversal of soft-tissue local anesthesia. In a previous study, this was achieved by injection of phentolamine mesylate, and novel LLLT therapy might be considered as a non-invasive treatment adjunct to local anesthesia [39,40].

In the present study, 635-nm and 808-nm diode lasers were used to accelerate the microcirculation and stop pulpal and soft-tissue anesthesia. Clinical studies about the effect of PBMT on pain reduction during local anesthesia injection in pediatric dentistry are still limited [41-44].

When the studies related to local anesthesia and PBM are examined, it is seen that some of them have been performed in adults and a few of them have been performed in children [24].

There are differences in research protocols in terms of sample size, anesthesia application region, laser type used, application method, and parameters. On the other hand, age was the most important criterion when assessing the self-inflicted injury risk.

There are many positive results of diode laser therapy on microcirculation that might have an impact on local anesthesia results [45]. The mechanism of PBM on vascular activity include increased level of endothelium-dependent vasodilator and improved vascular dysfunction in patients with diabetes, caused by the stimulating effect on collateral circulation and microcirculation in injured tissues [46].

Animal studies have demonstrated transcranial PBM can influence residual cerebral blood flow and decreased apoptosis of cells in the hippocampus [47,48]. Additionally, it has been demonstrated that photobiomodulation acts on pain via anti-inflammation effects, which might be considered as a secondary effect of PBM to achieve faster post-procedural recovery of sensory function, but many of its effects require time [49]. It has been reported in several studies that local anesthesia acts on neurotransmission and depolarization of the nerve membrane. El Feghali et al [50] reported that a wide range of wavelengths and parameters can induce analgesia. It has been shown that wavelengths of the visible and near-infrared spectra have different cellular targets. However, the main role of the mitochondria seems to be conserved and affected by a wide range of frequencies from visible to near-infrared. Consequently, at the basis of all these effects, modulation of ATP concentration can influence the action of Na+, K+-ATPase and its role in membrane depolarization, which supports the anti-nociceptive effect of drugs as well as neurotransmitter release [50]. Amaroli et al found that a diode laser at 808 nm can increase mitochondrial activities and release of specific neurotransmitters through membrane depolarization [51,52]. Both effects are involved in nerve pain management, helping to speed recovery of damage nerve function [53].

We found that irradiation with an 808-nm diode laser with energy density of 4J/cm^2 shortens the duration of soft-tissue anesthesia by about 30 min compared to the sham laser group. Similar effects were observed after irradiation of the anesthetized area with a 635-nm diode laser with energy density of 5J/cm^2. According to our results and previous studies, LLLT can increase microcirculation and blood flow in anesthetized areas, resulting in dilution of the anesthetic drug and more effective elimination of anesthetic effect.

In a group of children aged 3-15 years old, Podogrodzki et al found that application of a low-power 680-nm laser caused a significant increase in blood flow in the skin [53]. PBM performed after injection in our study resulted in a decreased duration of soft-tissue anesthesia by about 45 min. However, it was lower but acceptable when compared with the effect of phentolamine drug injection (40-70 min) reported in the literature [3]. It is also important that there is no need for reinjection of the drug and its associated complications, which influences children's behavior and cooperation during treatment.

Seraj et al investigated the clinical effect of photobiomodulation with use of an 810-nm diode laser on reversal of infiltration anesthesia in children. The study included 34 children 4-8 years old who needed infiltration anesthesia in the mandible. They found a significant difference in duration of soft-tissue anesthesia between the laser and sham laser groups (P<0.001). In conclusion, they proposed photobiomodulation with 810-nm diode laser as a non-invasive method for reversal of the effect of local anesthesia in children [24].

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Considering the need for dental services with less pain and more comfort, as well as the need for low-risk and non-pharmacological treatments, the present study tested PBM as an alternative to injections of phentolamine mesylate in acceleration of reversal of infiltration anesthesia performed in restorative procedures in children.

In summary, the vasodilator effect of PBM can lead to earlier removal of local anesthetic solution at the injection site, thus reducing the time to recovery of normal sensation. We found earlier recovery to normal pulp sensation after irradiation with 808-nm and 635-nm diode lasers. Clinically, we found that if the PBM is administered at 30 min, soft-tissue anesthesia duration was shorter compared with the sham group. However, pulpal anesthesia will also be affected for the remainder of the appointment. The early reversal would be most important in restorative dentistry, but may also have an effect on endodontic procedures when periapical anesthesia is lost. A limitation of the study was that the trial was not blind; however, the statistical power of 95% is an important strength. As a preliminary study, it should be continued with different parameters, such as different wavelengths and energy densities, and in younger children, who are the most susceptible to traumatic self-inflicted injuries after local anesthesia.

The major limitation in this study is that it only included healthy children who were 8-10 years old. Future research would ideally involve children younger than 8 years and special needs patients. Also, blinding of the examiner/investigator would strengthen the results. The potential of PBM to return patients' soft tissues to normal sensation and to avoid self-inflicted injury are highly desirable in pediatric dentistry and would be perceived as advantageous by many pediatric patients and their parents and/or guardians. The results revealed in our study establish an important clinical value for PBM in reducing the severity of postoperative injury and in accelerating recovery time to normal function from local anesthesia in pediatric patients after dental procedures.

The findings of this study demonstrate that after 15 min in the laser group, the percentage of individuals experiencing a return to normal sensations was significantly higher in the PBM groups compared to the control group. Specifically, 88% of individuals in the 808-nm PBM group and 68% in the 635-nm PBM group regained normal sensations, while only 20% of the control group did so. Additionally, after 45 min, all participants who received PBM reported a return to normal sensations. Furthermore, it was observed that both wavelengths of PBM resulted in a lower excitability threshold of the dental pulp compared to the control group. These findings provide evidence that PBM can effectively decrease the time required for individuals to recover normal sensations. It has been shown in this study that laser irradiation with 635-nm and 808-nm diode lasers can alleviate pulpal and soft tissues anesthesia, which was a predictor variable, and clinically diminishes the risk of self-inflicted injury without any adverse effects in children aged 8-10 years (primary and secondary endpoints). Innovation of the study is the use of PBM instead of phentolamine mesylate in cases when prolonged anesthesia can lead to self-injury.

Further research is needed to determine how different applications of laser parameters, injection techniques, and individual characteristics (eg, age) influence the effectiveness of PBM on acceleration of anesthesia effect. Additionally, PBM reduced the risk of postoperative self-inflicted soft-tissue injuries in children and increased their compliance with subsequent visits to pediatric dentists.

Conclusions

In the present study, PBM with the parameters described was effective and the time of reversal of the anesthetic effect was acceptable for patients. Our hypothesis was that PBM could be an alternative form of shortening the duration of local anesthesia and that it is as effective as the conventional injection of phentolamine mesylate. Thus, PBM might be considered a good alternative to the pharmacologic method of acceleration of reversal of soft-tissue anesthesia, especially as a non-invasive aid to prevent self-inflicted injury in children, as shown in our study.

Declaration of Figures' Authenticity

All figures submitted have been created by the authors who confirm that the images are original with no duplication and have not been previously published in whole or in part.

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